



Horizontal Asymptotes

Who Wins?

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* The Rules // "Who Wins?"

- Compare the Degrees

Compare	Who Wins?	HA
Bottom degree bigger	Bottom wins	$y = 0$
Same degree (TIE)	Tie $\rightarrow$ Ratio	$y = \frac{\text{lead top}}{\text{lead bottom}}$
Top degree bigger	Top wins	No HA

* Find the HA // Bottom Wins

$$f(x) = \frac{5}{x+3}$$

* Find the HA // Bottom Wins

$$f(x) = \frac{5}{x+3}$$

Top degree: **0** vs Bottom degree: **1**

Bottom wins!

HA: $y = 0$

* Find the HA // Bottom Wins

$$f(x) = \frac{3}{x^2 + 1}$$

* Find the HA // Bottom Wins

$$f(x) = \frac{3}{x^2 + 1}$$

Top degree: **0** vs Bottom degree: **2**

Bottom wins!

HA: $y = 0$

* Find the HA // Bottom Wins

$$f(x) = \frac{2x + 1}{x^2 - 9}$$

* Find the HA // Bottom Wins

$$f(x) = \frac{2x + 1}{x^2 - 9}$$

Top degree: 1 vs Bottom degree: 2

Bottom wins!

HA: $y = 0$

* Find the HA // Tie (Whole Number)

$$f(x) = \frac{4x}{x+2}$$

* Find the HA // Tie (Whole Number)

$$f(x) = \frac{4x}{x+2}$$

Top degree: **1** vs Bottom degree: **1** → TIE!

Leading coefficients: $\frac{4}{1} = 4$

HA: $y = 4$

* Find the HA // Tie (Whole Number)

$$f(x) = \frac{6x^2 - 1}{3x^2 + 5}$$

* Find the HA // Tie (Whole Number)

$$f(x) = \frac{6x^2 - 1}{3x^2 + 5}$$

Top degree: **2** vs Bottom degree: **2** → TIE!

Leading coefficients: $\frac{6}{3} = 2$

HA: $y = 2$

* Find the HA // Tie (Whole Number)

$$f(x) = \frac{9x}{3x + 4}$$

* Find the HA // Tie (Whole Number)

$$f(x) = \frac{9x}{3x + 4}$$

Top degree: **1** vs Bottom degree: **1** → TIE!

Leading coefficients: $\frac{9}{3} = 3$

HA: $y = 3$

* Find the HA // Tie (Fraction!)

Watch Out!

Sometimes the answer stays a **fraction**. That's okay!

$$f(x) = \frac{3x^2 + 1}{5x^2 - 2}$$

* Find the HA // Tie (Fraction!)

Watch Out!

Sometimes the answer stays a **fraction**. That's okay!

$$f(x) = \frac{3x^2 + 1}{5x^2 - 2}$$

TIE! Leading coefficients: $\frac{3}{5}$

$$\text{HA: } y = \frac{3}{5}$$

* Find the HA // Tie (Fraction!)

$$f(x) = \frac{5x - 10}{2x + 8}$$

* Find the HA // Tie (Fraction!)

$$f(x) = \frac{5x - 10}{2x + 8}$$

TIE! Leading coefficients: $\frac{5}{2}$

$$\text{HA: } y = \frac{5}{2}$$

* Find the HA // Tie (Fraction!)

$$f(x) = \frac{x-7}{x+4}$$

* Find the HA // Tie (Fraction!)

$$f(x) = \frac{x - 7}{x + 4}$$

TIE! Leading coefficients: $\frac{1}{1} = 1$

HA: $y = 1$

- **Pro Tip**

The leading coefficient of x is 1, even when it's not written!

* Find the HA // Tie (Negative!)

$$f(x) = \frac{-2x + 7}{6x - 1}$$

* Find the HA // Tie (Negative!)

$$f(x) = \frac{-2x + 7}{6x - 1}$$

TIE! Leading coefficients: $\frac{-2}{6} = -\frac{1}{3}$

$$\text{HA: } y = -\frac{1}{3}$$

Watch Out!

Don't forget the **negative sign!**

* Find the HA // Top Wins

$$f(x) = \frac{x^2 + 1}{x - 3}$$

* Find the HA // Top Wins

$$f(x) = \frac{x^2 + 1}{x - 3}$$

Top degree: 2 vs Bottom degree: 1

Top wins!

No HA

* Find the HA // Top Wins

$$f(x) = \frac{x^3 + 2}{x^2 - 1}$$

* Find the HA // Top Wins

$$f(x) = \frac{x^3 + 2}{x^2 - 1}$$

Top degree: **3** vs Bottom degree: **2**

Top wins!

No HA

* Your Turn! // What's the HA?

▷ *YouTry!*

$$f(x) = \frac{x+1}{x^3-8}$$

Top degree: _____ vs Bottom degree: _____

Who wins? _____

HA: _____

* Your Turn! // What's the HA?

▷ *YouTry!*

$$f(x) = \frac{x+1}{x^3-8}$$

Top degree: _____ vs Bottom degree: _____

Who wins? _____

HA: _____

Bottom wins (1 vs 3) → **HA: $y = 0$**

* Your Turn! // What's the HA?

▷ *You Try!*

$$f(x) = \frac{x^2 - 4x}{2x^2 + 3x}$$

Top degree: _____ vs Bottom degree: _____

Who wins? _____

HA: _____

* Your Turn! // What's the HA?

▷ *YouTry!*

$$f(x) = \frac{x^2 - 4x}{2x^2 + 3x}$$

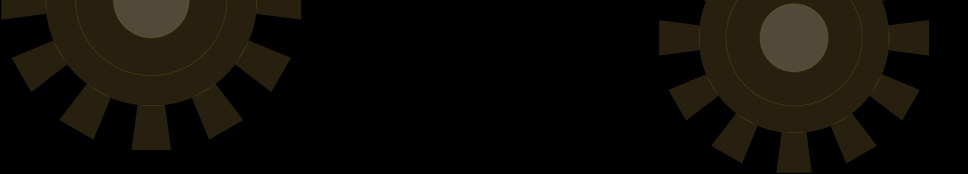
Top degree: _____ vs Bottom degree: _____

Who wins? _____

HA: _____

Tie! $\frac{1}{2}$ →

$$\text{HA: } y = \frac{1}{2}$$



Go Sandies!



You know who wins.